

STAT 210A Final Cheat sheet

Loss Function: $L(\theta, d)$, measures disutility of guessing $g(\theta)=d$
Risk Function: $R(\theta; \delta(\cdot)) = \mathbb{E}_\theta[L(\theta, \delta(X))]$ ↑ Convexity Loss means in terms of d

↳ **Ex:** risk for square error loss is $MSE(\theta; \delta(\cdot)) = \mathbb{E}_\theta[(\delta(X) - g(\theta))^2]$
Inadmissibility: δ inadmissible if $\exists \delta^*$ st. $\begin{cases} R(\theta; \delta^*) \leq R(\theta; \delta) \text{ for all } \theta \\ R(\theta; \delta^*) < R(\theta; \delta) \text{ for some } \theta \end{cases}$
 ↳ δ^* strictly dominates δ

Average-Case Risk: $\int_\Theta R(\theta; \delta) d\pi(\theta)$ for some prior measure π
 If π is prob. meas. $\Rightarrow \mathbb{E}_{\theta \sim \pi}[R(\theta; \delta)]$ (minimizing results in Bayes Estimator)

Worst-Case Risk: $\sup_\theta R(\theta; \delta)$ (minimizing results in Minimax Estimator)

Exponential Families

s-param Expo Family: $\mathcal{P} = \{P_n : n \in \Xi\}$ w/ densities p_n of form:
 $p_n(x) = e^{n^T T(x) - A(n)} h(x)$,

↳ **convex!** where $T: \mathcal{X} \rightarrow \mathbb{R}^S$ sufficient statistic
 $h: \mathcal{X} \rightarrow \mathbb{R}$ carrier/base density
 $n \in \Xi \subseteq \mathbb{R}^S$ natural parameter
 $A: \mathbb{R}^S \rightarrow \mathbb{R}$ log-partition function

↳ Can interpret as tilting $h(x)$ towards $T(x)$ (via $e^{n^T T(x)}$) and then renormalizing via $e^{-A(n)}$. ↳ convex b/c $A(n)$ convex!

Natural Parameter Space: $\Xi = \{n : A(n) < \infty\}$, normalizable p_n

Differential Identities: $A(n)$ has all partial derivatives on Ξ , by Keener Thm 2.4

(1st) $\frac{\partial A}{\partial n_j}(n) = \mathbb{E}_n[T_j(X)]$, $\nabla A(n) = \mathbb{E}_n[T(X)] \in \mathbb{R}^S$
 (2nd) $\frac{\partial^2 A}{\partial n_j \partial n_k}(n) = \text{Cov}_n(T_j, T_k)$, $\nabla^2 A(n) = \text{Var}_n(T(X)) \in \mathbb{R}^{S \times S}$ ↳ $J(n) = \frac{1}{n} J(\theta)$

Moment Generating Function: $M_n^T(u) = e^{A(n+u) - A(n)}$ (MGF of $T(X)$ when $X \sim P_n$)
 ↳ k th order moment of $T(X) = \frac{\partial^k}{\partial n^k} (e^{A(n)}) \cdot \frac{1}{e^{A(n)}}$

Cumulant Generating Function: $K_n^T(u) = \log M_n^T(u) = A(n+u) - A(n)$ (CGF of $T(X)$)
 Model w/ $p_\theta(x)$ wrt meas. μ

Sufficiency

↳ **Definition:** $T(X)$ sufficient for $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ if $P_\theta(x|T)$ independent of θ

Sufficiency Principle: if $T(X)$ sufficient for $\mathcal{P}$, then any stat procedure should depend on X only thru $T(X)$. $\theta \rightarrow T(x) \xleftrightarrow{X} \tilde{x}$ (just as good as X)

Factorization Theorem: $T(X)$ sufficient for model $\mathcal{P}$ iff $\exists g_\theta(t), h(x)$ st.

$p_\theta(x) = g_\theta(T(x)) h(x)$
 for μ -almost-every x : $\mu(\{x : p_\theta(x) \neq g_\theta(T(x)) h(x)\}) = 0$

Order Statistics: $(X_{(i)})_{i=1}^n$ ($X_{(k)}$ is k th smallest) is sufficient

Empirical Distribution: $\hat{P}_n(\cdot) = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}(\cdot)$ is sufficient, where $\delta_{x_i}(A) = 1\{x_i \in A\}$

Minimal Sufficiency: $T(X)$ minimal sufficient if (1) $T(X)$ sufficient; and
 (2) $\forall$ other sufficient $S(X)$, $T(X) = f(S(X))$ for some f .

↳ *If $T(X)$ sufficient and $T(X) = f(S(X))$, then $S(X)$ sufficient.

Likelihood Function: $Lik(\theta; X) = p_\theta(x)$

↳ **Log-likelihood:** $\ell(\theta; x) = \log Lik(\theta; x)$

↳ Likelihood up to scaling (or ℓ up to vert shift) is minimal sufficient!

↳ Likelihood ratios $\left(\frac{Lik(\theta_1; x)}{Lik(\theta_2; x)}\right)_{\theta_1, \theta_2 \in \Theta}$

Minimal Sufficiency via Likelihood: $T(X)$ minimal sufficient if

(1) $T(X)$ sufficient
 (2) $T(X)$ can be recovered from likelihood shape
 $Lik(\cdot; x) \propto Lik(\cdot; y) \Rightarrow T(x) = T(y)$
 or equivalently,
 $\ell(\cdot; x) - \ell(\cdot; y) = \text{const}(x, y) \Rightarrow T(x) = T(y)$

Minimal Sufficiency of Expo Fam: $T(X)$ minimal if $\text{Span}\{\eta(\theta_1) - \eta(\theta_2) : \theta_1, \theta_2 \in \Theta\} = \mathbb{R}^S$

Completeness

Strat for completeness proofs: show two things are a.s. equal by showing they have equal expectation

↳ **Definition:** $T(X)$ is complete for $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ if, for any measurable f ,
 $\mathbb{E}_\theta[f(T(X))] = 0 \forall \theta \Rightarrow f(T) \stackrel{a.s.}{=} 0 \forall \theta$

Full-rank Expo Fam: $\mathcal{P}$ is full-rank if $T(X)$ satisfies no linear constraint
 $(\nexists \beta \neq 0, \alpha : \beta^T T(X) \stackrel{a.s.}{=} \alpha)$, and Ξ contains an open set.

↳ If $\mathcal{P}$ is not full-rank, we say it is **curved**

↳ If $T(X)$ satisfies linear constraint, $\mathcal{P}$ might still be full-rank for lower-dim suff. stat.

Theorem: If $\mathcal{P}$ full-rank, then $T(X)$ is complete sufficient

Theorem: If $T(X)$ complete sufficient for $\mathcal{P}$, then $T(X)$ minimal

Ancillarity

↳ **Definition:** $V(X)$ ancillary for $\mathcal{P}$ if its distro independent of θ

Basu's Theorem: If $T(X)$ complete sufficient and $V(X)$ ancillary for $\mathcal{P}$, then
 $V(X) \perp\!\!\!\perp T(X)$ for all $\theta \in \Theta$

↳ If can't verify Thm's hypotheses for one family, try another!

Unbiased Estimation

↳ **Require:** $\mathbb{E}_\theta[\delta(X)] = g(\theta) \forall \theta \in \Theta$. If $T(X)$ complete sufficient, then there exists at most one unbiased $\delta^*(T(X))$, which uniformly minimizes risk for any convex loss function

Jensen's: If f convex then $f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)] \forall$ r.v. X

Rao-Blackwell Theorem: $T(X)$ sufficient, $\delta(X)$ estimator, $\tilde{\delta}(T(X)) = \mathbb{E}[\delta(X) | T(X)]$

↳ **Recipe to improve any $\delta(X)$ that violates suff. principle**
 Then, we have: $L(\theta, d)$ convex $\Rightarrow R(\theta, \tilde{\delta}) \leq R(\theta, \delta)$ ↳ No θ !
 $L(\theta, d)$ strictly convex $\Rightarrow R(\theta, \tilde{\delta}) < R(\theta, \delta)$ (unless $\delta(X) \stackrel{a.s.}{=} \tilde{\delta}(T(X)) \forall \theta$)

* $\tilde{\delta}(T) = \text{Rao-Blackwellization of } \delta(X)$

U-estimable: $g(\theta)$ U-estimable if $\exists \delta(X)$ st. $\mathbb{E}_\theta[\delta(X)] = g(\theta) \forall \theta$

Uniform Min Variance Unbiased (UMVU): $\delta(X)$ UMVU if $\forall$ unbiased $\tilde{\delta}$,
 $\text{Var}_\theta(\delta(X)) \leq \text{Var}_\theta(\tilde{\delta}(X)) \forall \theta \in \Theta$
Use power series on $\mathbb{E}_\theta \delta(X)$ and match terms

Finding UMVUE: 1) Find any unbiased estimator based on complete suff. $T(X)$
 2) Find any unbiased estimator at all, then R-B'ize it (i.e. $\tilde{\delta}(T)$)

Score → **Definition:** $\nabla \ell(\theta; X)$, thought of as "local complete suff. stat."

Differential Identities: $\mathbb{E}_\theta[\nabla \ell(\theta; X)] = 0$, $\text{Var}_\theta[\nabla \ell(\theta; X)] = \mathbb{E}_\theta[-\nabla^2 \ell(\theta; X)]$ Note all θ same
No unbiased estimator can have smaller variance than $\nabla g(\theta)' J(\theta)^{-1} \nabla g(\theta)$

Cramer-Rao Lower-Bound: Let $g(\theta) = \mathbb{E}_\theta[\delta(X)]$, then:
 1-param: $\text{Var}_\theta(\delta) \cdot \text{Var}_\theta(\dot{\ell}(\theta; x)) \geq \text{Cov}_\theta(\delta, \dot{\ell}(\theta; x)) \Rightarrow \text{Var}_\theta(\delta) \geq \frac{g(\theta)^2}{J(\theta)}$

Multivar: $\theta \in \mathbb{R}^q$, $g(\theta), \delta(X) \in \mathbb{R}$: $\text{Var}_\theta(\delta) \geq \nabla g(\theta)' J(\theta)^{-1} \nabla g(\theta)$

Efficiency: $\text{eff}_\theta(\delta) = \frac{\text{CRLB}}{\text{Var}_\theta(\delta)} \leq 1$, say $\delta(X)$ **efficient** if $\text{eff}_\theta(\delta) = 1 \forall \theta$
(of unbiased est) $\text{Corr}_\theta^2(\delta, \dot{\ell}(\theta; x))$ $\text{Corr}_\theta^2(\delta, \dot{\ell}(\theta)) = 1 \forall \theta$

Bayes Estimation

Bayes Risk: $r_\Delta(\delta) = \int_\Theta R(\theta; \delta) d\Delta(\theta) = \mathbb{E}_{\theta \sim \Delta}[R(\theta; \delta)] = \mathbb{E}_{(\theta, x)}[L(\theta, \delta(x))]$

Densities: prior $\lambda(\theta)$, likelihood $p_\theta(x)$, joint density $\lambda(\theta)p_\theta(x)$,
 marginal density $q(x) = \int_\Theta \lambda(\theta)p_\theta(x) d\theta$, posterior density $\lambda(\theta|x) = \frac{\lambda(\theta)p_\theta(x)}{q(x)}$

Bayes Estimator: $\delta_\Delta(x) \in \arg \min \mathbb{E}[L(\theta, d) | X=x]$ a.e. (w.p. 1)

↳ **Ex:** if $L(\theta, d) = (g(\theta) - d)^2$ then $\delta_\Delta(x) = \mathbb{E}[g(\theta) | X=x]$

↳ **Ex:** if $L(\theta, d) = w(\theta)(g(\theta) - d)^2$ then $\delta_\Delta(x) = \frac{\mathbb{E}[w(\theta)g(\theta) | X]}{\mathbb{E}[w(\theta) | X]}$

Conjugate Prior: prior is conjugate to likelihood if posterior from same family

Likelihood	Prior
$X_i \theta \sim \text{Binomial}(n, \theta)$	$\theta \sim \text{Beta}(\alpha, \beta)$
$X_i \theta \sim N(\theta, \sigma^2)$	$\theta \sim N(\mu, \tau^2)$
$X_i \theta \sim \text{Pois}(\theta)$	$\theta \sim \text{Gamma}(v, s)$

Empirical Bayes: Common in HBM

$\xi \sim \lambda(\xi)$, $\theta_i | \xi \stackrel{iid}{\sim} \pi_\xi(\theta)$, $X_i | \xi, \theta \stackrel{iid}{\sim} p_\theta(x)$

↳ **Hybrid Approach:** treat ξ as fixed
 1. Estimate ξ based on observed data
 2. Plug in ξ as though it were known

Hierarchical Bayes

Ex. Predict a batter's "true" batting average from n_i at-bats. $X_i = \# \text{ of hits} \sim \text{Bin}(n_i, \theta_i)$
 Pool info across players via **hierarchical model**: $\alpha, \beta \sim \text{Dir}(\alpha, \beta)$, $\theta_i | \alpha, \beta \sim \text{Beta}(\alpha, \beta)$, $X_i | \theta_i \sim \text{Bin}(n_i, \theta_i)$
 $\mathbb{E}[\theta_i | X] = \mathbb{E}[\mathbb{E}[\theta_i | X, \alpha, \beta] | X] = \mathbb{E}[\frac{X_i + \alpha}{n_i + \alpha + \beta} | X]$ $\leftarrow$ fixed $\leftarrow$ sampled $\sim \pi(\alpha, \beta | X)$
 $\tau^2 \sim \lambda_0$, $\theta_i | \tau^2 \sim N(0, \tau^2)$, $X_i | \tau^2, \theta_i \sim N(\theta_i, 1)$
Gaussian Hierarchical Model: $\delta(X_i) = \mathbb{E}[\mathbb{E}[\theta_i | X, \tau^2] | X] = \mathbb{E}[\tau^2 / (1 + \tau^2) \cdot X_i | X] = \mathbb{E}[\tau^2 / (1 + \tau^2) | X] \cdot X_i$
Linear Shrinkage Estimator $\rightarrow \delta(X_i) = (1 - \mathbb{E}[\tau^2 | X]) X_i$ where $\tau^2 = 1 / (1 + \tau^2) \leftarrow$ shrinkage factor

Markov Chain Monte Carlo

$\lambda(\theta | x) = \frac{p(\theta, x)}{\int p(\theta, x) d\theta}$ $\leftarrow$ nice to compute $\leftarrow$ often intractable
 $\rightarrow$ Compute Strat. use MC w/ stationary dist $\alpha p_\theta(x | \theta)$, run it to approximate samples from $\lambda(\theta | x)$
 $\rightarrow$ Definition: sequence of rvs $X^{(0)}, X^{(1)}, \dots$ st. $X^{(0)} \sim p_\theta, X^{(t+1)} | X^{(t)} \sim Q(\cdot | X^{(t)})$
 where $Q(y | x) = P[X^{(t+1)} = y | X^{(t)} = x]$
Stationary Distribution: $\pi(y) = \int Q(y | x) \pi(x) dx$ $\leftarrow$ MC satisfying DBE are reversible
 $\rightarrow$ Detailed Balance: $\pi(x) Q(y | x) = \pi(y) Q(x | y) \forall x, y$ (sufficient for stationarity)

Gibbs Sampler

Given param vector $\theta = (\theta_1, \dots, \theta_d)$
 $\rightarrow$ Algorithm: Initialize $\theta = \theta^{(0)}$
 for $t = 1, \dots, T$:
 for $j = 1, \dots, d$:
 Sample $\theta_j \sim \pi(\theta_j | \theta_{-j}, X)$
 Record $\theta^{(t)} = \theta$
 Variations:
 • Update one random coord $J^{(t)} \sim \text{Unif}\{1, \dots, d\}$
 • Update coords in random order
 $\rightarrow \propto P(\theta_j | \theta_{-j}, X) \prod_{i \in \mathcal{I}(\theta_j)} P(\theta_i | \theta_{-i}, X)$
 Properties: • If $\theta^{(t)} \sim \lambda(\theta | x)$ then $\theta^{(t+1)} \sim \lambda(\theta | x)$
 $\rightarrow$ Updating any coord in any order preserves posterior distro

James-Stein Estimator

Some Definitions: $(Dh(x))_{ij} = \frac{\partial^2 h(x)}{\partial x_i \partial x_j}$, $\|A\|_F = (\sum_{i,j} A_{ij}^2)^{1/2}$
 Estimator: $\delta_{JS, d}(X) = (1 - \frac{d-2}{\|X\|^2}) X_i$ for $d \geq 3$ $\leftarrow$ Empirical Bayes Motivation: $\frac{d-2}{\|X\|^2}$ is UMVUE of $\frac{1}{\|X\|^2}$
 Paradox: $X_i \sim N(\theta_i, \sigma^2 I_d)$, $\theta \in \mathbb{R}^d$ (fixed), $\sigma^2 > 0$ known. Then, $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is inadmissible as estimator of θ under squared error loss ($\delta_{JS}(X)$ better)
 Lemma: Assume $X \sim N_d(\theta, \sigma^2 I_d)$, $h: \mathbb{R}^d \rightarrow \mathbb{R}^d$, $\mathbb{E}[\|Dh(x)\|_F^2] < \infty$: $\mathbb{E}[(X - \theta)' h(X)] = \sigma^2 \mathbb{E}[\text{tr}(Dh(x))] = \sigma^2 \sum_{i=1}^d \mathbb{E}[\frac{\partial h_i}{\partial x_i}(x)]$
 SURE: $\hat{R}(x) = \sigma^2 d + \|h(x)\|^2 + 2\sigma^2 \text{tr}(Dh(x))$ (unbiased estimator of MSE of any $\delta(x)$)
 $\uparrow h(x) = X - \delta(x)$

Minimax Risk

$\rightarrow$ Definition: $r^* = \inf_{\delta} \sup_{\theta} R(\theta; \delta)$
 Minimax Estimator: $\delta^* \in \arg \inf_{\delta} \sup_{\theta} R(\theta; \delta)$ (i.e. achieves minimax risk)
 Least Favorable Prior: Δ^* where $r_{\Delta^*} = \sup_{\theta} R(\theta; \delta^*)$, $\sup_{\theta} R(\theta; \delta) \geq r^* \geq r_{\Delta^*} \geq r_{\Delta}$
 Theorem: if $r_{\Delta} = \sup_{\theta} R(\theta; \delta_{\Delta})$ w/ Bayes δ_{Δ} , then:
 (1) δ_{Δ} minimax
 (2) Unique Bayes $\Rightarrow$ Unique minimax
 (3) Δ least favorable
 $\rightarrow$ avg risk = sup risk if $R(\theta; \delta_{\Delta})$ const and $\Delta(\{\theta: R(\theta; \delta_{\Delta}) = \max_{\theta'} R(\theta'; \delta_{\Delta})\}) = 1$
 Least Favorable Sequence: sequence $\{\Delta_n\}_{n=1}^\infty$ is LF if $r_{\Delta_n} \rightarrow \sup_{\theta} R(\theta; \delta)$
 Bounds: Upper: $r^* \leq \sup_{\theta} R(\theta; \delta)$ $\forall \delta$
 Lower: $r^* \geq \int R(\theta; \delta_{\Delta}) d\Delta(\theta) \forall \theta$

Hypothesis Testing

Null: $H_0: \theta \in \Theta_0 \subseteq \Theta$
 Alt: $H_1: \theta \in \Theta_1 \subseteq \Theta$
 Rejection region: $R = \{x: \phi(x) = 1\}$
 Acceptance region: $A = \{x: \phi(x) = 0\}$
 Critical/Test Function: $\phi(x) = \begin{cases} 1 & \text{accept } H_1 \\ 0 & \text{reject } H_0 \end{cases}$ $\rightarrow \phi$ is level- α test (a.e. $\theta \in \Theta_0$) if $\sup_{\theta \in \Theta_0} \mathbb{E}_{\theta}[\phi(x)] \leq \alpha$
 Power Function: $\beta_{\phi}(\theta) = \mathbb{E}_{\theta}[\phi(x)] = P_{\theta}[\text{Reject } H_0]$
 Likelihood Ratio Test (LRT): $\phi^*(x) = \begin{cases} 1 & \text{if } \frac{LR(x)}{LR(x_0)} > c \\ 0 & \text{if } \frac{LR(x)}{LR(x_0)} \leq c \end{cases}$ where c, γ chosen to make $\mathbb{E}_{\theta_0}[\phi^*(x)] = \alpha$
 $\rightarrow$ Neyman Pearson Lemma: level- α LRT is optimal for testing $H_0: X \sim p_0$ vs. $H_1: X \sim p_1$ (simple hypotheses)

UMP

$\phi^*(x)$ is uniformly most powerful if for any other same-lvl test ϕ we have $\mathbb{E}_{\theta}[\phi^*(x)] \geq \mathbb{E}_{\theta}[\phi(x)] \forall \theta \in \Theta$
 Monotone Likelihood Ratios: $P = \{p_{\theta}: \theta \in \Theta \subseteq \mathbb{R}^d\}$ has MLR if $\exists T(x)$ st. $\frac{p_{\theta_1}(x)}{p_{\theta_2}(x)}$ is a non-decreasing function of $T(x)$ for any $\theta_1 < \theta_2$
 Theorem: Assume P has MLR, test $H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0$ at level $\alpha \in (0, 1)$. Let $\phi^*(x) = \begin{cases} 1 & \text{if } T(x) < c \\ 0 & \text{if } T(x) \geq c \end{cases}$ $\leftarrow$ c, γ chosen st. $\mathbb{E}_{\theta_0}[\phi^*(x)] = \alpha$
 ϕ^* is LRT $\rightarrow$ Then (1) ϕ^* is a UMP level- α test; and (2) If $\theta_0 < \theta_1$ then ϕ^* minimizes $\mathbb{E}_{\theta_1}[\phi(x)]$ among all ϕ st. $\mathbb{E}_{\theta_0}[\phi(x)] = \alpha$

One-Sided Test

$H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0$, often no UMP test exists
 Stochastically Increasing: $T(x)$ is SI in θ if $P_{\theta}[T(x) \geq t]$ is non-increasing in θ , $\forall t$
 $\rightarrow$ If $\phi(x)$ right-tailed on $T(x)$ and $T(x)$ SI, then $\mathbb{E}_{\theta}[\phi(x)]$ is increasing in θ
 Two-Sided Test: extreme values of $T(x)$: $\phi(x) = \begin{cases} 1 & \text{if } T(x) \leq c_1 \text{ or } T(x) \geq c_2 \\ 0 & \text{if } T(x) \in (c_1, c_2) \end{cases}$

Equal-Tailed Test

Let $\alpha_1 = P_{\theta_0}[T(x) < c_1]$, $\alpha_2 = P_{\theta_0}[T(x) > c_2]$, $\alpha = \alpha_1 + \alpha_2$
 $\alpha_1 = P_{\theta_0}[T(x) < c_1] = \alpha/2$, $\alpha_2 = P_{\theta_0}[T(x) > c_2] = \alpha/2$

Unbiased Test

$\phi(x)$ unbiased if $\inf_{\theta \in \Theta_0} \mathbb{E}_{\theta}[\phi(x)] \geq \alpha$ (ensure min $\beta(\theta) = \alpha$) $\rightarrow$ Choose c_1, γ and c_2, γ_2 to solve:
 $\beta(\theta_0) = \alpha$, $\frac{d\beta(\theta)}{d\theta}(\theta_0) = 0$

Theorem

Assume $X_i \sim \text{iid } e^{\theta T(x) - A(\theta)} h(x)$ and two-tailed test. Then:
 (1) Unbiased test on $\Sigma T(x_i)$ w/ level α is UMPU (UMP among all unbiased tests)
 (2) If $\theta < \theta_0$, UMPU test found by solving c_1, γ_1 st. $\mathbb{E}_{\theta_0}[\phi] = \mathbb{E}_{\theta}[\phi] = \alpha$
 (3) If $\theta > \theta_0$, UMPU test found by solving c_2, γ_2 st. $\beta_{\theta_0}(\theta_0) = \alpha$ and $\frac{d\beta_{\theta_0}(\theta)}{d\theta}(\theta_0) = \mathbb{E}_{\theta_0}[\Sigma T(x_i) (\phi(x) - \alpha)] = 0$

p-value

Informal Definition: $p(x) = P_{H_1}[T(x) \geq T(x)] = \sup_{\theta \in \Theta_1} P_{\theta}[T(x) \geq T(x)]$
 Assume we have ϕ_{α} for each α , $\sup_{\theta \in \Theta_0} \mathbb{E}_{\theta}[\phi_{\alpha}(x)] \leq \alpha$, and tests are monotone in α . Then:
 Definition: $p(x) = \inf \{\alpha: \phi_{\alpha}(x) = 1\}$ ($= \inf \{\alpha: x \in R_{\alpha}\}$ for non-randomized)
 $\rightarrow$ For $\theta \in \Theta_0$, $P_{\theta}[p(x) \leq \alpha] \leq \inf_{x \in R_{\alpha}} P_{\theta}[\phi_{\alpha}(x) = 1] \leq \alpha \Rightarrow$ p-value stochastically dominates $\text{Unif}[0, 1]$

Confidence Regions

$C(X)$ is a 1- α confidence set for $g(\theta)$ if coverage probability $= P_{\theta}[C(X) \ni g(\theta)] \geq 1 - \alpha, \forall \theta \in \Theta$ $\leftarrow$ C(X) random, not $g(\theta)$
 $\rightarrow C(X)$ covers $g(\theta)$ if $C(X) \ni g(\theta)$; Confidence Level: $\inf_{\theta \in \Theta} P_{\theta}[C(X) \ni g(\theta)]$
 Duality of Tests and Confidence Sets: (1) Given level- α test $\phi(x, \alpha)$ of $H_0: g(\theta) = a$ vs. $H_1: g(\theta) \neq a, \forall a \in g(\Theta)$. Then, we can construct CS for $g(\theta)$: $C(X) = \{a: \phi(x, \alpha) < 1\}$
 $\rightarrow P_{\theta_0}[C(X) \ni g(\theta_0)] = P_{\theta_0}[\phi(x, \alpha) = 1] \geq 1 - \alpha \forall \theta_0$
 (2) Given 1- α CS $C(X)$ for $g(\theta)$. Then, can construct test $\phi(x)$ of $H_0: g(\theta) = a$ vs. $H_1: g(\theta) \neq a, \phi(x) = 1$ if $a \notin C(X)$.
 $\rightarrow \mathbb{E}_{\theta_0}[\phi(x)] = P_{\theta_0}[C(X) \ni g(\theta_0)] \leq \alpha \forall \theta_0$ st. $g(\theta) = a$
 $\rightarrow$ Procedure in (1) is called "Inverting the test"

Confidence Interval: A CS $C(X)$ is a CI if it's in the form $C(X) = [C_1(X), C_2(X)]$
 $\rightarrow$ Lower Confidence Bound $LCB(X) = [C_1(X), \infty)$
 $\rightarrow$ Upper Confidence Bound $UCB(X) = (-\infty, C_2(X)]$
 $\rightarrow$ Obtained by inverting a one-sided test
 $\rightarrow$ If test UMP, called Uniformly Most Accurate (UMA)
 $\rightarrow$ Obtain CI by inverting two-sided test; if UMPU then called UMPU

Nuisance Parameters

Setup: $P = \{P_{\theta, \lambda}: (\theta, \lambda) \in \Omega\}$, $H_0: \theta \in \Theta_0$ vs. $H_1: \theta \in \Theta_1$
 Parameter of Interest $\rightarrow$ Nuisance Parameter: extra unknown param that might affect stuff
 Multiparameter: Assume $X \sim p_{\theta, \lambda}(x) = \exp\{\theta' T(x) + \lambda' U(x) - A(\theta, \lambda)\}$ where $\theta \in \mathbb{R}^d, \lambda \in \mathbb{R}^r$ both unknown
 Exp. Families: • To test $H_0: \theta \in \Theta_0$ vs. $H_1: \theta \in \Theta_1$, we perform the following: * Idea: condition on $U(X)$ to elim. λ
 1. Sufficiency Reduction: $(T(X), U(X)) \sim q_{\theta, \lambda}(t, u) = \exp\{\theta' t + \lambda' u - A(\theta, \lambda)\} g(t, u)$
 w/ $c(u), \gamma(u)$ chosen st. $\mathbb{E}_{\theta, \lambda}[\phi^*(X)] = \alpha$
 2. Condition on $U(X)$: $q_{\theta}(t | u) = q_{\theta, \lambda}(t, u) / \int q_{\theta, \lambda}(t, u) d\lambda = \exp\{\theta' t - B_{\theta}(u)\} g(t | u)$ where $B_{\theta}(u) = \int \lambda' u g(t | u) d\lambda$
 3. Conditional Test: Test $H_0: \theta \in \Theta_0$ vs. $H_1: \theta \in \Theta_1$ in model $\mathcal{Q}_u = \{q_{\theta}(t | u): \theta \in \Theta\}$
 Note: if $s=1$, $\mathcal{Q}_u$ has MLR in T
 Theorem: 1) For $H_0: \theta \in \Theta_0$ vs. $H_1: \theta \in \Theta_1$, use UMPU $\phi^*(X) = \psi(T(X); U(X))$ where $\psi(t; u) = \begin{cases} 1 & t > c(u) \\ 0 & t \leq c(u) \end{cases}$
 w/ $c(u), \gamma(u)$ chosen st. $\mathbb{E}_{\theta_0}[\phi^*(X) | U(X)=u] = \alpha, \mathbb{E}_{\theta_0}[\psi(T(X); U(X)) | U(X)=u] = 0$
 2) For $H_0: \theta = \theta_0$ vs. $H_1: \theta \neq \theta_0$, use UMPU $\phi^*(X) = \psi(T(X); U(X))$ where $\psi(t; u) = \begin{cases} 1 & t \in [c_1(u), c_2(u)] \\ 0 & t \in (-\infty, c_1(u)) \cup (c_2(u), \infty) \end{cases}$
 w/ $c_1(u), \gamma_1(u)$ chosen st. $\mathbb{E}_{\theta_0}[\phi^*(X) | U(X)=u] = \alpha, \mathbb{E}_{\theta_0}[\psi(T(X); U(X)) | U(X)=u] = 0$

Change of Basis

Asymptotics: $X_n \xrightarrow{P} c$ if $P[\|X_n - c\| > \epsilon] \rightarrow 0, \forall \epsilon > 0$
 $X_n \xrightarrow{L} X$ if $\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)] \forall$ bdd, cts $f: X \rightarrow \mathbb{R}$
 Consistency: $\delta_n(X_n)$ consistent for $g(\theta)$ if $\delta_n(X_n) \xrightarrow{P} g(\theta)$, i.e. $P_{\theta}[\|\delta_n(X_n) - g(\theta)\| > \epsilon] \rightarrow 0$
 LLN: if $\mathbb{E}[X_i] < \infty$, $\mathbb{E}[X_i^2] < \infty$, then $\bar{X}_n \xrightarrow{P} \mu, \bar{X}_n^2 \xrightarrow{P} \mu^2$
 CLT: if $\mathbb{E}[X_i] = \mu \in \mathbb{R}^d, \text{Var}(X_i) = \Sigma < \infty$, then $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \Sigma)$
 Delta Method: if $\sqrt{n}(\bar{X}_n - \mu) \Rightarrow N_d(0, \Sigma)$ and $f: \mathbb{R}^d \rightarrow \mathbb{R}^k$ diff at μ , then $\sqrt{n}(f(\bar{X}_n) - f(\mu)) \Rightarrow N_k(0, Df(\mu) \Sigma Df(\mu)')$
 $\rightarrow$ In general, can do higher-order Taylor expansion: if $f(\mu) = 0$, we have
 $n(f(\bar{X}_n) - f(\mu)) \Rightarrow \bar{f}(\mu) \sqrt{n}(\bar{X}_n - \mu) \approx \bar{f}(\mu) \sigma \sqrt{2} \cdot \chi^2_k$
 $Df(x) = \begin{pmatrix} -\nabla f_1(x) \\ \vdots \\ -\nabla f_k(x) \end{pmatrix}$

Maximum Likelihood Estimation

$\rightarrow$ Definition: for dominated $P = \{P_{\theta}: \theta \in \Theta\}$, MLE is $\hat{\theta}_{MLE}(X) = \arg \max_{\theta \in \Theta} p_{\theta}(X) = \arg \max_{\theta \in \Theta} \ell(\theta; X)$
 Asymptotic Efficiency: Assume $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_{\theta}(x)$, where p_{θ} smooth in θ . Let: $\ell_1(\theta; X_i) = \log p_{\theta}(X_i)$
 $J_1(\theta) = \text{Var}_{\theta}(\nabla \ell_1(\theta; X_i)) = -\mathbb{E}_{\theta}[\nabla^2 \ell_1(\theta; X_i)]$
 $J_n(\theta) = \text{Var}_{\theta}(\nabla \ell_n(\theta; X)) = n J_1(\theta)$
 Then, estimator $\hat{\theta}_n$ is asymptotically efficient if $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{P} N(0, J_1(\theta)^{-1})$
 $\rightarrow$ Delta Method for $g(\theta)$: $\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{P} N(0, \nabla g(\theta)' J_1(\theta)^{-1} \nabla g(\theta))$
 $\rightarrow \hat{\theta}_{MLE}$ is asymptotically gaussian and efficient
 KL-Divergence: $D_{KL}(\theta_1 || \theta_2) = \mathbb{E}_{\theta_1}[\log p_{\theta_1}(X)/p_{\theta_2}(X)]$
 Asymptotic Distro of MLE: Assume $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_{\theta_0}$ for $\theta_0 \in \Theta \subseteq \mathbb{R}^d$, $\hat{\theta}_n \in \arg \max_{\theta \in \Theta} \ell_n(\theta; X)$, $\hat{\theta}_n \rightarrow \theta_0$, etc., and $\mathbb{E}_{\theta_0}[\nabla \ell_1(\theta_0; X)] = 0$, $\text{Var}_{\theta_0}[\nabla \ell_1(\theta_0; X)] = -\mathbb{E}_{\theta_0}[\nabla^2 \ell_1(\theta_0; X)] > 0$. Then we have:
 $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{P} N_d(0, J_1(\theta_0)^{-1})$

Likelihood-Based Inference

Some assumptions as (*) directly above. We then have $\sqrt{n} \nabla \ell_n(\hat{\theta}_n) \Rightarrow N_d(0, J_1(\theta_0))$, $\sqrt{n} \nabla^2 \ell_n(\hat{\theta}_n) \Rightarrow J_1(\theta_0)$, $\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N_d(0, J_1(\theta_0)^{-1})$
 Wald-Type Confidence: Assume estimator $\hat{\theta}_n \rightarrow \theta_0$ st. $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{P} N_d(0, \Sigma)$. Then, to test $H_0: \theta = \theta_0$ vs. $H_1: \theta \neq \theta_0$ we reject if $\|\sqrt{n}(\hat{\theta}_n - \theta_0)\|^2 > \chi^2_d(\alpha)$. This gives confidence ellipsoid.
 $\rightarrow$ Option 1: $\hat{\theta}_n = n J_1(\hat{\theta}_n) n \text{Var}_{\hat{\theta}_n}(\nabla \ell_n(\hat{\theta}_n; X))$
 $\rightarrow$ Option 2: $\hat{\theta}_n = -\nabla^2 \ell_n(\hat{\theta}_n; X) \leftarrow$ Preferred! Univariate Wald Interval: $\theta_j \in C_j = \hat{\theta}_{nj} \pm \sqrt{(\hat{\theta}_n^{-1})_{jj}} \cdot z_{\alpha/2}$
 Score Test: For $H_0: \theta = \theta_0$ vs. $H_1: \theta \neq \theta_0$, use score as test stat; then $J_n(\hat{\theta}_n)^{-1/2} \nabla \ell_n(\hat{\theta}_n; X) \xrightarrow{P} N_d(0, I_d)$.
 $\rightarrow$ We reject if $\|\sqrt{n}(\hat{\theta}_n - \theta_0)^{-1/2} \nabla \ell_n(\hat{\theta}_n; X)\|^2 > \chi^2_d(\alpha)$ * Asymptotically same as Wald Test
 $\rightarrow$ No quadratic approx, no MLE, no need to estimate Fisher info at θ_0
 GLRT w/ simple null: for $H_0: \theta = \theta_0$ vs. $H_1: \theta \neq \theta_0$, use test stat $2(\ell_n(\hat{\theta}_n; X) - \ell_n(\theta_0; X)) \Rightarrow \chi^2_d$
 GLRT w/ composite null or nuisance params: use thm to get $2(\ell_n(\hat{\theta}_n) - \ell_n(\hat{\theta}_0)) \Rightarrow \chi^2_{d-d_0}$
 Asymptotic Relative Efficiency: $\hat{\theta}_n^{(1)}$ w/ $i=1, 2$ estimators w/ det and $\sqrt{n}(\hat{\theta}_n^{(1)} - \theta_0) \Rightarrow N(0, \sigma_1^2)$, then ARE of $\hat{\theta}_n^{(2)}$ wrt $\hat{\theta}_n^{(1)}$ is σ_1^2 / σ_2^2
 $\hat{\theta}_0 = \arg \max_{\theta \in \Theta} \ell_n(\theta; X)$

Miscellaneous

• Bias $\alpha(\delta(X)) = \mathbb{E}_{\theta_0}[\delta(X)] - \theta$, MSE $\alpha(\delta; S) = \mathbb{E}_{\theta_0}[\|\delta(X) - \theta\|^2] = \text{Bias}\alpha(\delta(X))^2 + \text{tr}(\text{Var}_{\theta_0}(\delta(X)))$
 • For expo family, MLE sets complete sufficient statistic equal to expectation and solves for θ Model Identifiable: $P_{\theta} \neq P_{\theta_0}$ for $\theta \neq \theta_0$.
 Consistency of MLE: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_{\theta}, \theta \in \Theta$. Assume (1) p_{θ} cts in θ ; (2) Θ compact; (3) $\mathbb{E}_{\theta_0}[\sup_{\theta \in \Theta} |w_i(\theta)|] < \infty$; (4) Model Identifiable.
 Then, $\hat{\theta}_n \xrightarrow{P} \theta_0$ if $\hat{\theta}_n \in \arg \max_{\theta \in \Theta} \ell_n(\theta; X)$